



A.M.($\leq$)	Any	Moon P.	Gray
Image Quality		Seeing < 1.3"	
Cloudy Cover		Photometric	

Title of Proposal: Obtaining rotation light curves of Neptune and Uranus Trojans to determine their 3D shapes

Abstract:

Neptune and Uranus Trojans are small bodies located at the Lagrangian points of this gas giants. Not much is known about these objects, but it's believed that they were captured during the early planetary migration, keeping characteristics of the early Solar System. In this project, we propose a detailed and systematic determination of the shape, size, rotation period and pole orientation of 2013KY18 (Neptune Trojan located at L5), 2010EN65 (Neptune Trojan located at L3) and 2011QF99 (Uranus Trojan located at L4) using SOAR rotation light curves, stellar occultation events and Vera Rubin Observatory data. This is a pioneer project of Neptune and Uranus Trojans shape determination using these techniques. The results will make us better understand the collisional past and relations between different dynamical classes of Solar System small bodies.

Time requested: 24 hours.

Minimum time accepted: 12 hours.

Observing Mode: Classical Remote (CR)

Program Type: Regular Proposal (RP)

Principal Investigator (PI): Eros de Oliveira Gradovski

PI Affiliation: ON

PI Phone/PI Email: 41995435898 erosgradovski@gmail.com

Co-authors

Name	Affiliation	E-mail
Felipe Braga Ribas	UTFPR/DAFIS	fribas@utfpr.edu.br
Giuliano Margoti	ON	giulianomargoti@gmail.com
Julio Ignacio Bueno De Camargo	ON	camargo@on.br
Marcelo Assafin	OV/UFRJ	massaf@ov.ufrj.br
Bruno Eduardo Morgado	OV/UFRJ	bmorgado@ov.ufrj.br
Altair Ramos Gomes-junior	UFU	altairgomesjr@gmail.com
Roberto Vieira-Martins	ON	rvm@on.br
Bruno Sicardy	Observatoire de Paris	bruno.sicardy@obspm.fr
Jose Luis Ortiz	IAA/CSIC	ortiz@iaa.es
Josselin Desmars	Observatoire de Paris	josselin.desmars@obspm.fr
Chrystian Luciano Pereira	ON	chryslp.fis@gmail.com
Rafael Sfair	FEG/UNESP	rafael.sfair@unesp.br
Jonatã Arcas Silva	ON	jonatasilva@on.br
Tiago Francisco Lins Leal Pinheiro	ON	franciscopinheiro@on.br
Luan Catani	OV/UFRJ	lcatani@ov.ufrj.br
Viviane Peixoto	OV/UFRJ	vivianepeixoto989@gmail.com
Helio Dutra	OV/UFRJ	hdutra@ov.ufrj.br

Is there a thesis or dissertation that will benefit from the data?: YES

Student	E-mail	Degree	Defense
Eros de Oliveira Gradoski	erosgradovski@gmail.com	PhD student	03/2029
Giuliano Margoti	giulianomargoti@gmail.com	PhD student	11/2028

Instruments:

Instrument	Camera
Goodman (Imaging)	Red Camera

Instrument	Filter	Grating	Slit
Goodman (Imaging)			
	$\tilde{g}$ SDSS 4x4		
	$\tilde{r}$ SDSS 4x4		
	$\tilde{i}$ SDSS 4x4		

Optimal dates:

February 15 to 20, April 15 to 20, June 13 to 17

Impossible dates:

January 28 to February 8, February 25 to March 9, March 26 to April 8, April 23 to May 9, May 24 to 9 June, June 22 to

Scientific Justification

The study of small bodies provides crucial information to understand our Solar System formation and evolution. These objects have some regions of concentrations in our system, such as the main belt asteroid, Kuiper belt, and at the planet's Lagrangian points of equilibrium. A lot is known about Jupiter Trojans (JTs), since a large number of objects had already been observed at this region, but not much is known about Neptune Trojans (NTs) and Uranus Trojans (UTs). Previous works suggest that the NTs region could have a similar or even bigger number of objects when compared to JTs [1]. Similar to what is proposed for JTs, it's believed that the NTs and UTs were captured during the early planetary migration [2] [3] [4], having characteristics in their shape and color about their primitive formation region. The formation region is still an unsolved problem, where a possible source could be the Kuiper's belt. Recent works compared the color distribution of JTs, NTs and Kuiper belt objects searching for a similarity, however the results showed that the Kuiper belt objects had a red ($g-r < 0.75$) and ultra red ($g-r > 0.75$) color, but for NTs no ultra red had been observed so far [7]. For JTs only a few ultra reds had been found, which might be explained due to thermal resurfacing [8], as these bodies are closer to the Sun, which could have modified their original characteristics. However, this explanation fails for Neptune Trojans, as the low temperatures at their distance make thermal resurfacing unlikely [7].

A possible solution for NTs bluer surface would be a collisional resurface, where collisions would reveal a bluer interior as the ultra red surface would be ripped from interactions. One suggested source for these interactions would be the plutinos in a 3:2 resonance with Neptune that overlap with the Neptune Trojans region [9]. For UTs only 2 objects at this region had been found [10] [11], where almost nothing is known about their surface composition, size, and shape.

Giving this context studying the composition, size, and shape of these objects provide information about their formation origin and collisional past. For this reason, we separated 2 Neptune Trojans 2013KY18 (located at L5) and 2010EN65 (located at L3), and one Uranus Trojan 2011QF99 (located at L4) that are promising targets of new stellar occultations at this and the next 2 years. Our objective is to combine the occultation results that provide a precise view of the object shape at a 2D plane during the occultation event, with rotation light curves obtained at this proposal in order to obtain a 3D model of these objects. As the authors of this proposal are also members of the Brazilian Participation Group at the Legacy Survey of Space and Time project (BPG-LSST), the magnitude over time data obtained at Vera Rubin Observatory will also be integrated to generate the model.

To obtain the object's shape, the light curve inversion technique [12] will be applied through the All-Data Asteroid Modeling software [13]. The observation of a rotation light curves with SOAR for these objects will provide information of their rotation period, pole orientation and restrict their shapes. To compare an initial shape and the real data, a synthetic light curve using a Lambert Lommel-Seelinger scattering law [12] is generated, and a Root Mean Squared test between the observed and synthetic data is performed. The synthetic curves will also be generated for the Vera Rubin Observatory data and the stellar occultation results for the 2D view of the body will be compared to the model's limb, resulting in a scaled model with a shape that reproduces all data and provides the rotation period and pole orientation within their uncertainties (Figure 1).

The rotation observation at different filters will also provide information about these object's colors, where combined with their 3D shape, could indicate a collisional past (irregular shapes), that promotes the idea of a collisional resurface to explain missing ultra red Trojans. The images taken at SOAR will also be used for astrometry propose, providing a smaller uncertainty at the prediction of the stellar occultation events.

This proposal is a pioneering project aimed at a detailed study of the shape and size of Neptune and Uranus Trojans through rotational light curves and stellar occultations, providing a better understanding of these dynamical classes.

References

- [1] Lin, H. Chen, Y. Volk, K., et al., 2021, *Icarus*, 361, 114391.
- [2] Kortenkamp, S. Malhotra, R. & Michtchenko, T., 2004, *Icarus*, 167, 347.
- [3] Tsiganis, K. Gomes, R. Morbidelli, A. et al., 2005 *Nature*, 435, 459.
- [4] Nesvorný, D. & Vokrouhlický, D., 2009 *AJ*, 137, 5003.
- [5] Sheppard, S., 2010, *AJ*, 139, 1394.

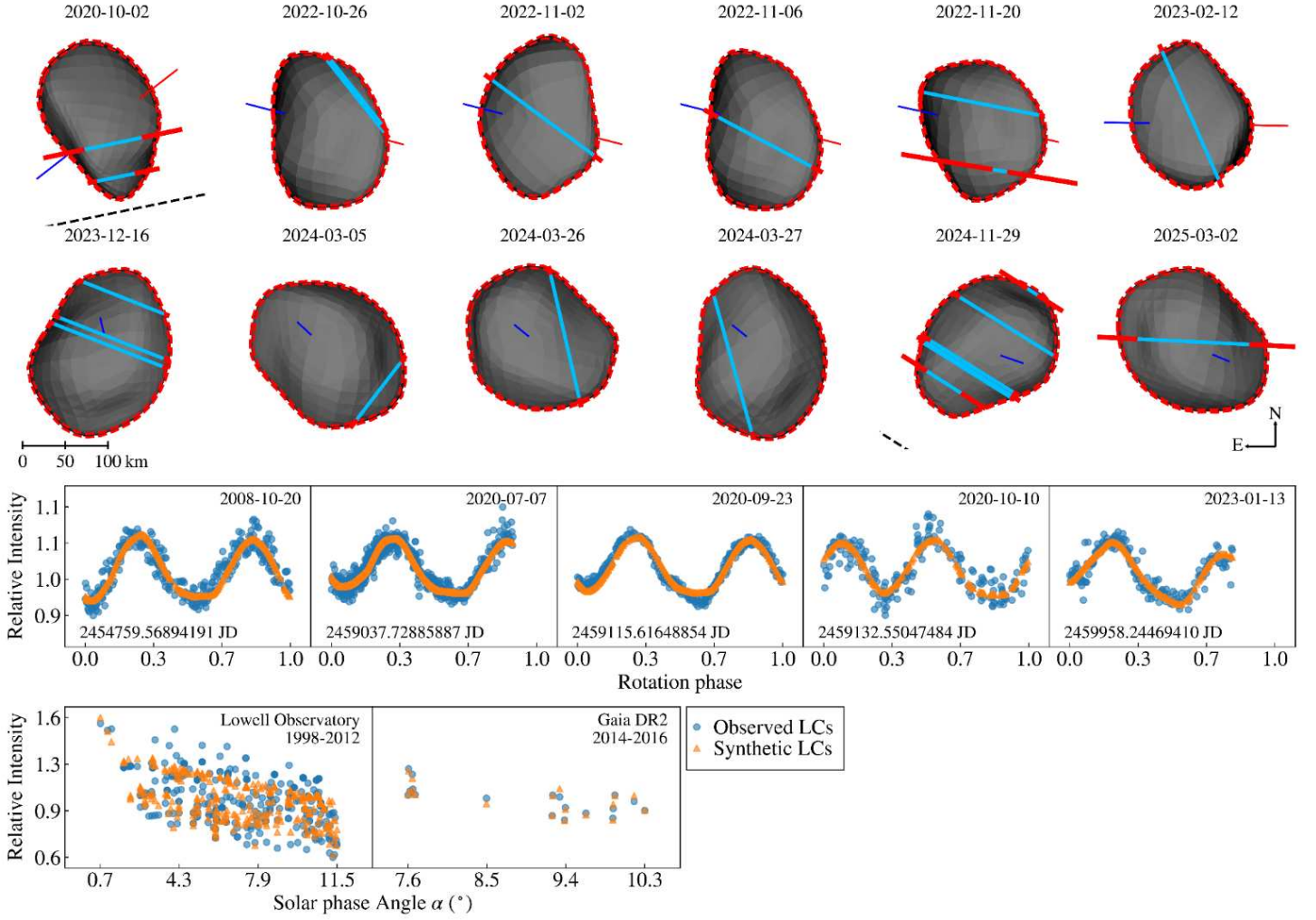


Figure 1: Example of a 3D model generated for a Jupiter Trojan by the PI of this project using rotation light curves, solar phase angle curves and stellar occultation data. The final solution represents all data simultaneously. At the top of the figure we can see the stellar occultation events, where each blue line with a red bar at the extremities represent a different observation of a sized section of the body within its uncertainties (red part). The darker blue line and red line coming from the center of the object represent its pole direction during the event. The bottom part of the figure shows a comparison between the observed data and a synthetic light curve generated when rotating the model, considering the obtained rotation period and pole orientation.

- [6] Schwamb, M. Fraser, W. Bannister, M. et al., 2019 ApJS, 243, 12.
- [7] Jewitt, D., 2018, AJ, 155, 56.
- [8] Jewitt, D., 2002, AJ, 123, 1039.
- [9] Almeida, A. Peixinho, N., & Correia, A., 2009, A&A, 508, 1021.
- [10] Marcos, C. & Marcos, R., 2017, MNRAS, 467, 1561.
- [11] Alexandersen, M., 2013, Science, 341, 997.
- [12] Kaasalainen, M. Torppa, J. & Muinonen, K., 2001, Icarus, 153, 37.
- [13] Viikinkoski, M. Kaasalainen, M. & Āurech, J., 2015, A&A, 576, A8.

Experimental Design

The Neptune Trojans 2013KY18, 2010EN65 and the Uranus Trojan 2011QF99 were selected as they are promising targets for future stellar occultation events and are possible to be observed at SOAR. For the Neptune Trojans its specially interesting that 2013KY18 is a stable Trojan located at L5, and 2010EN65 is a transitory Trojan located at L3 that might become a future Centaur (objects that orbit between Jupiter and Neptune), so the comparison of their characteristics might show interesting results.

For all three objects, only orbital proprieties are available in the literature, so there is no information about their rotation period or pole direction. An estimate of the size of these objects was made assuming an absolute magnitude, where 2013KY18 would have a diameter of 241 km, 2010EN65 with 204 km and 2011QF99 with 62 km¹. We request 24 hours of observation in total, where 8 hours would be used for each object. If possible, it's important that the 8 hours for each object is at the same night of observation, allowing us to cover as much of its rotation as possible in case the object has a long rotation period.

In order to get the best possible data, we need nights that the targets are far from the moon or with no moon, as the apparent magnitude V of these targets are around 22. Our objective is to measure their flux variation during the night and compare with other reference stars at the field. Since we will use this data to generate the 3D model, a higher number of exposures results in a more detailed shape, as we cover more parts of the body rotating. Our goal is to use the smallest exposure possible, where we estimate that a 450 seconds exposure time would be needed to achieve an $\text{SNR} > 50$, resulting in a precision of 0.02 magnitude. The traditional dark and flat images will also be needed in order to calibrate the scientific images before the photometry.

The resulting images will be analyzed using the Package for the Reduction of Astronomical Images Automatically (PRAIA)² that performs a differential aperture photometry. The resulting light curve will be used at the All-Data Asteroid Modeling (ADAM)³ software with the stellar occultation and Vera Rubin Observatory data to derive the 3D model, rotation period and pole orientation of these objects. The final model will be submitted to a Python code that variates these parameters within small increments in order to precisely obtain their uncertainties.

The photometric process with PRAIA of SOAR's images will also be used to obtain new astrometric positions of the targets, reducing the uncertainty at the stellar occultation events predictions, making them more likely to be observed (Figure 2).

¹Johnston, W. List of known trans-Neptunian objects. Available at: <https://www.johnstonsarchive.net/astro/tnoslist.html>. Accessed: October 10, 2025.

²ASSAFIN, M., 2023, Planetary and Space Science, 239, 105816.

³Viikinkoski, M. Kaasalainen, M. & Āurech, J., 2015, A&A, 576, A8.

Technical Description

At this project we pretend to use the Goodman (Imaging) Red Camera in the R filter to obtain the rotation light curves as it offers a good flux and image quality. The g' , r' , and i' SDSS filters will also be used for some images in order to obtain the object's color index.

The 8 hours pretended for each of the 3 objects (totalizing 24 hours) has to be disposed at separate dates as each object has a better observation window at different epochs:

- For 2011QF99 (Uranus Trojan, L4) the best dates are from February 15 to 20, where the object is going to be observable from the astronomical twilight to ~ 4 am (UTC) (Figure 3) and with low or no moon contamination. In this case, the first half (4 hours) of 2 nights would be ideal, since the object is not visible throughout the entire night.
- For 2010EN65 (Neptune Trojan, L3) the best observing dates are from April 15 to 20, where the object is observable from the sun set until the sun rise (Figure 4), and considering moon conditions. In this case, a complete night (8 hours) would be ideal.
- For 2013KY18 (Neptune Trojan, L5) the best observation window is from June 13 to 17, where the object will also be observable for 8 hours (Figure 5). Since in June the nights are longer, the extra hour at the beginning of the night could be used to observe 2010EN65 before 2013KY18 is observable.

In addition to the 24 hours requested for observation, 2 hours will also be necessary for overheads, such as flat and dark calibration images.

For all proposed dates, the target fields were analyzed to ensure that no nearby stars would contaminate the acquisition of the targets' rotation light curves. The expected apparent magnitude for 2011QF99, 2010EN65 and 2013KY18 are 22.2, 21.9 and 21.4 respectively at the intended dates. For this reason, as mention above, we expect to use an exposure time between 350 and 450 seconds depending on the quality of the night.

Previous missions of this proposal to this telescope

This proposal has not been submitted before.

Previous missions of other proposals to this telescope

Other proposals of rotation light curves, but for Centaurs and Trans Neptunian objects, this would be the first for Neptune and Uranus Trojans:

- "Chiron's rotation light curve for 3D shape characterization" (SO2023B-017): four half-nights of data obtained. The results are in analysis. PI: Chrystian Luciano Pereira
- "Rotation light curve for the high-inclination TNO 2004XR190" (SO2021B-012): six half-nights of data obtained. Results from these observations were recently presented at the XLV Meeting of the Brazilian Astronomical Society (2022). PI: Flavia Luane Rommel

Previous results in the field by the Principal Investigator

- Gradovski, E., Obtenção dos parâmetros rotacionais e escala de troianos de Júpiter através de curvas de rotação e ocultações. Dissertação (Mestrado em Física e Astronomia) - Universidade Tecnológica Federal do Paraná, Curitiba, 2025.
- Gradovski, E. et al., Stellar occultations by (3451) Mentor, an irregular Trojan. In: XLV Reunião Anual da Sociedade Astronômica Brasileira, 2023. Proceedings da XLV Reunião Anual da SAB. v. 34. p. 80-81.
- Rizos, J. et al., The trans-Neptunian object (119951) 2002 KX14 revealed via multiple stellar occultations, A&A 697 A62 (2025), DOI: 10.1051/0004-6361/202554154.
- Dutra H. et al., 2025, Physical characteristics of Jupiter's Trojan (1437) Diomedes from a tri-chord stellar occultation in 2020 and dimensionless three-dimensional model, Phil. Trans. R. Soc. A.38320240187 DOI: 10.1098/rsta.2024.0187

PI and Co-Is publications relevant to this proposal

Related works that evolve stellar occultations or rotation light curves:

- Pereira, C. et al., Physical properties of Centaur (60558) 174P/Echeclus from stellar occultations, Monthly Notices of the Royal Astronomical Society, Volume 527, Issue 2, January 2024, Pages 3624–3638.
- Sicardy, B. et al., Stellar occultations by trans-Neptunian objects. Astron Astrophys Rev 32, 6 (2024).
- Gomes-Júnior, A. et al., The first observed stellar occultations by the irregular satellite Phoebe (Saturn IX) and improved rotational period, Monthly Notices of the Royal Astronomical Society, Volume 492, Issue 1, February 2020, Pages 770–781.
- Morgado, B. E., et al., Refined physical parameters for Chariklo's body and rings from stellar occultations observed between 2013 and 2020. Astronomy & Astrophysics, v. 652, p. A141, 2021.
- Braga-Ribas, Felipe et al. Constraints on (2060) Chiron's size, shape, and surrounding material from the November 2018 and September 2019 stellar occultations. Astronomy & Astrophysics, v. 676, p. A72, 2023.
- Morgado, B. E., et al., A stellar occultation by the transneptunian object (50000) Quaoar observed by CHEOPS. Astronomy & Astrophysics. v.664, p.L15, 2022.
- Morgado, B. E., et al., A dense ring of the trans-Neptunian object Quaoar outside its Roche limit. Nature. v.614, p.239 - 243, 2023.

Software develop by the Co-Is that will be used to perform the analysis of this project:

- Assafin, M. Differential aperture photometry and digital coronagraphy with PRAIA. Planetary and Space Science, v. 239, p. 105816, 2023.
- Gomes-Júnior, A. R. et al., SORA: Stellar occultation reduction and analysis. Monthly Notices of the Royal Astronomical Society, v. 511, n. 1, p. 1167-1181, 2022.

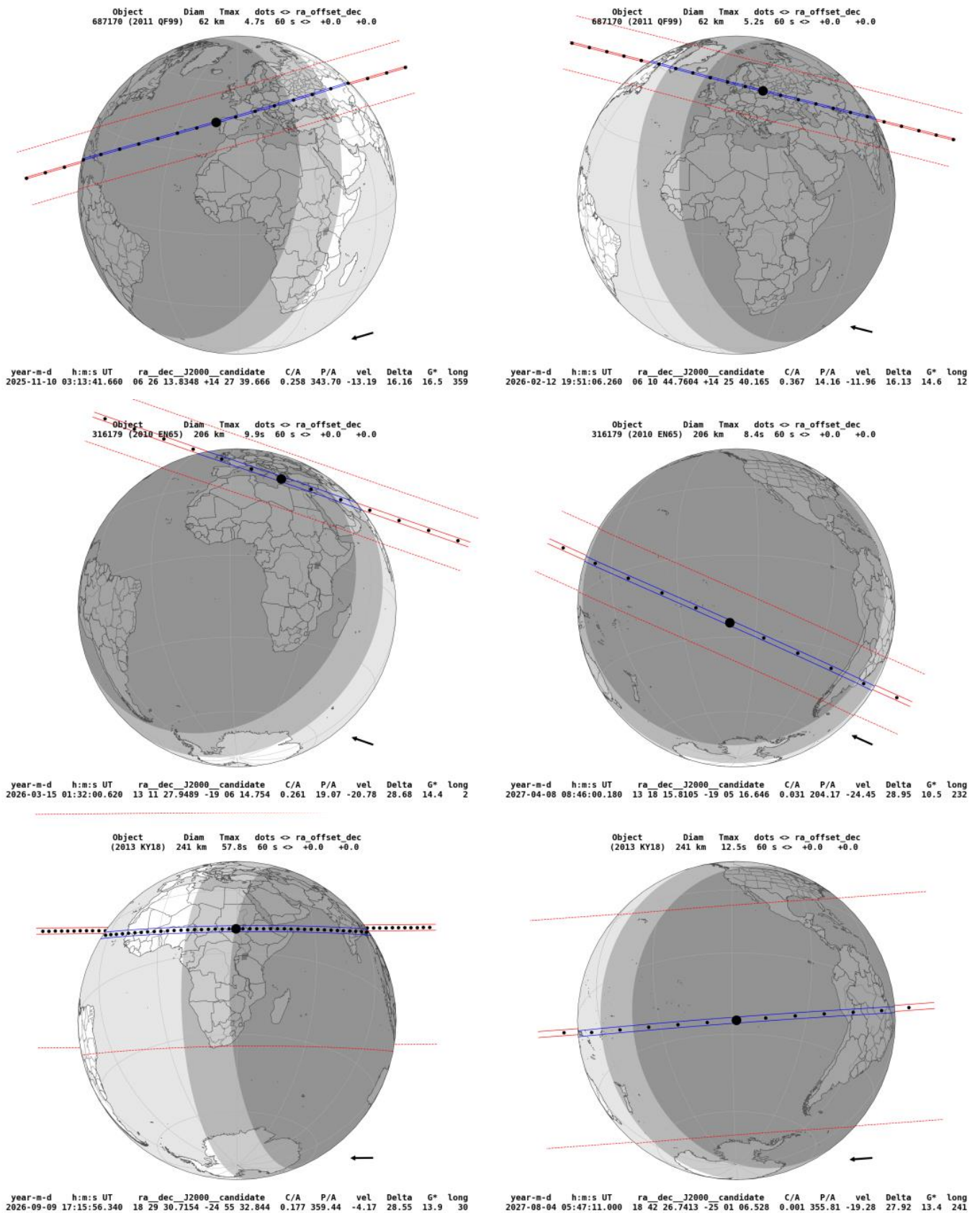


Figure 2: Example of some predicted occultations for 2013KY18, 2010EN65 and 2011QF99. The continued parallel blue lines represents the object shadow projected at Earth's surface, and the red dashed lines represent the prediction's uncertainty.

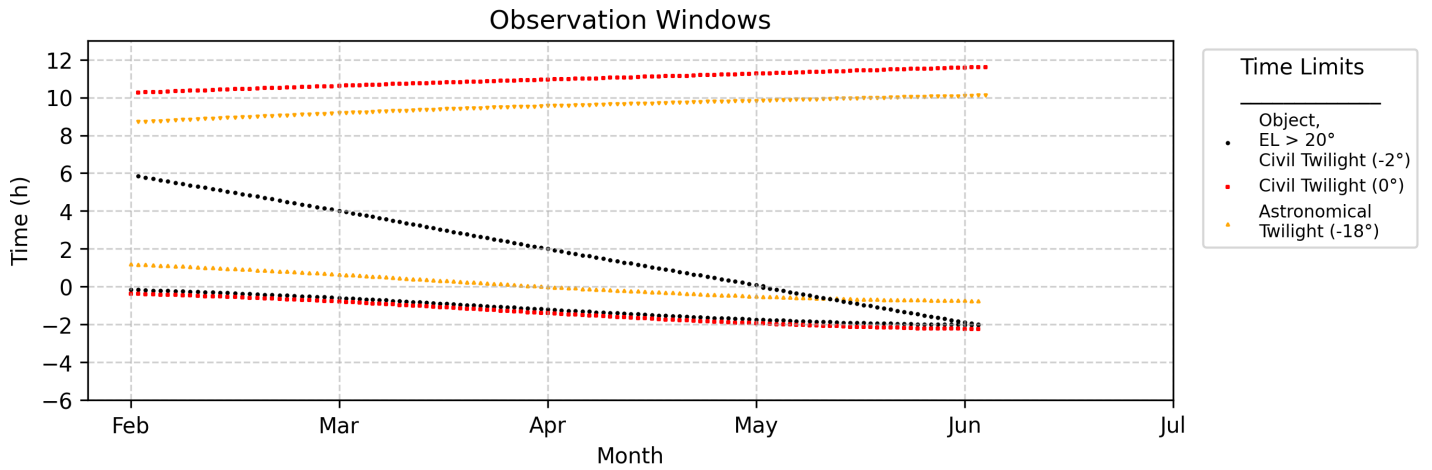


Figure 3: Observation condition for 2011QF99 (Uranus Trojan, L4). The red dotted lines represents the civil twilight, the yellow lines the astronomical twilight and the black lines represents the interval in which the object has an elevation $> 20^\circ$.

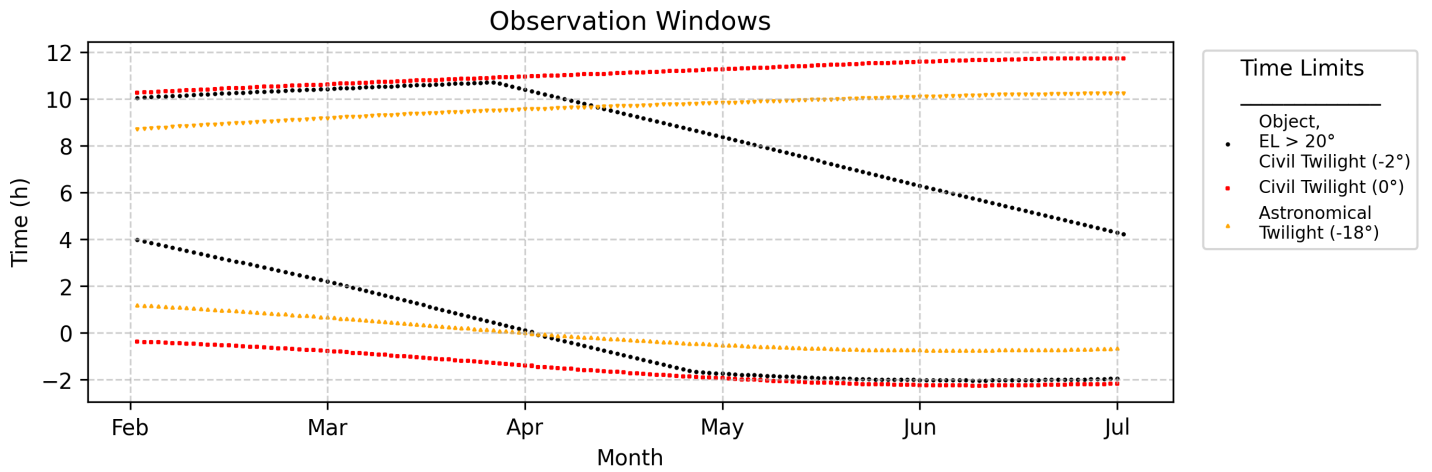


Figure 4: Observation condition for 2010EN65 (Neptune Trojan, L3). (See description Figure 3).

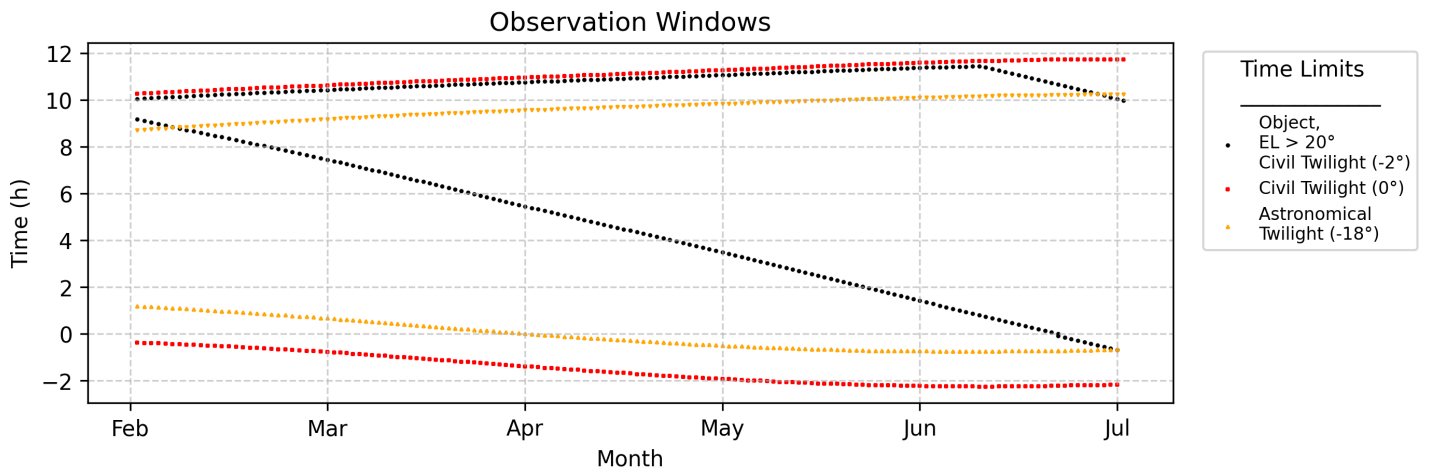


Figure 5: Observation condition for 2013KY18 (Neptune Trojan, L5). (See description Figure 3).

Objects List.

Object	A.R.(h,m)	Dec.(deg)	Mag.	Band	Type	Comments
2011QF99	06:10:32.02	+14:26:30.4	22.2	V	Sci	at 2026-Feb-15 02:00
2010 EN65	13:08:11.86	-18:46:34.0	21.9	V	Sci	at 2026-Apr-15 02:00
2013KY18	18:37:50.14	-24:47:08.4	21.4	V	Sci	at 2026-Jun-13 03:00